

4.5 The Dimensions of a Vector Space

I

Theorem 9 in Section 4.4 implies that a vector space V with a basis B containing n vectors is isomorphic to $\mathbb{R}^n$.

This section shows that this number n is an intrinsic property (called the dimension) of the space V that does not depend on the particular choice of basis.

The discussion of dimension will give additional insight into properties of bases. The first theorem generalizes a well-known result about the vector space $\mathbb{R}^n$.

Theorem 10

If a vector space V has a basis $B = \{\vec{b}_1, \dots, \vec{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

p.265

Theorem 11

If a vector space V has a basis of n vectors, then every basis of V must consist of exactly n vectors.

p.265

If a nonzero vector space V is spanned by a finite set S , then a subset of S is a basis for V , by the spanning set theorem (p.249, theorem 4.3.5). In this case, theorem 11 ensures that the following definition makes sense.

Definition

If a vector set V is spanned by a finite set, then V is said to be finite-dimensional, and the dimension of V , written as $\dim V$, is the number of vectors in a basis for V .

The dimension of the zero vector space $\{0\}$ is defined to be zero. If V is not spanned by a finite set, then V is said to be infinite-dimensional.

p.266

Example 1

The standard basis for $\mathbb{R}^n$ contains n vectors, thus $\dim \mathbb{R}^n = n$. The standard polynomial basis $\{t^0, t^1, t^2\}$ shows that $\dim \mathbb{P}_2 = 3$. In general, $\dim \mathbb{P}_n = n+1$.

The space $\mathbb{P}$ of all polynomials is infinite-dimensional.

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II

Example 2

Let $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$, where $\vec{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

Then H is the plane studied in Example 7 in section 4.4. A basis for H is $\{\vec{v}_1, \vec{v}_2\}$, since $\vec{v}_1$ and $\vec{v}_2$ are not multiples and hence are linearly independent. Thus $\dim H = 2$.

Clearly, $\vec{v}_1 \neq \vec{0}$, $\vec{v}_2$ is not a multiple of $\vec{v}_1$, but $\vec{v}_3$ is a multiple of $\vec{v}_2$. By the spanning set theorem, we may discard $\vec{v}_3$ and still have a set that spans H . Finally, $\vec{v}_4$ is not a linear combination of $\vec{v}_1$ and $\vec{v}_2$. So $\{\vec{v}_1, \vec{v}_2, \vec{v}_4\}$ is linearly independent (by Theorem 4 in section 4.3 p.247) and hence is a basis for H . Thus, $\dim H = 3$.

Example 3

Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a-3b+6c \\ 5a+4d \\ b-2c-d \\ 5d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}$$

Solution

Lets put it in vector form

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \vec{v}_4 = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

We notice that $\vec{v}_3$ is a multiple of $\vec{v}_2$.

Example 4

The subspaces of $\mathbb{R}^3$ can be classified by dimension.

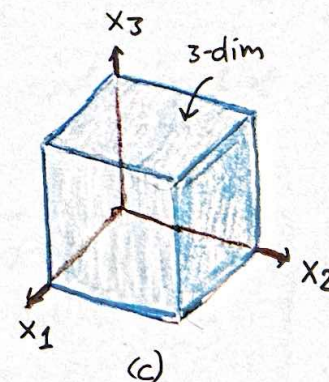
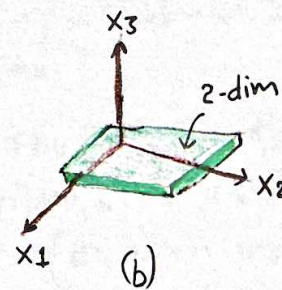
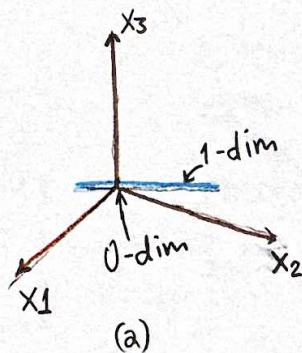


Figure 1- Sample subspaces of $\mathbb{R}^3$.

4.5 Dimensions of a Vector Space (cont'd)

III

Subspaces of a Finite-Dimensional Space

The next theorem is a natural counterpart to the spanning set theorem (p. 249).

Theorem 12

Let H be a subspace of a finite-dimensional vector space V . A linearly independent set in H can be expanded, if necessary, to a basis for H . Also, H is finite-dimensional and

$$\dim H \leq \dim V$$

p. 267

Theorem 13

Let V be a p -dimensional, $p \geq 1$. Any linearly independent set of exactly p elements in V is automatically a basis for V . Any set of exactly p elements that spans V is automatically a basis for V .

p. 267

The Dimensions for Nul A , Col A , and Row A

Since the dimensions of the null space and the column space of an $m \times n$ matrix are referred to frequently, they have specific names:

Definition

The rank of an $m \times n$ matrix A is the dimension of the column space and the nullity of A is the dimension of the null space.

The rank of an $m \times n$ matrix A is the number of pivot columns and the nullity of A is the number of free variables. Since the dimension of the row space is the number of pivot rows, it is also equal to the rank of A .

Putting these observations together results in the rank theorem.

Theorem 14 - The Rank Theorem

The dimensions of the column space and the null space of an $m \times n$ matrix A satisfy the equation

$$\text{rank } A + \text{nullity } A = \text{number of columns in } A$$

p. 268

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Example 5

Find the nullity and rank of

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & -2 & 2 & 3 & -1 \\ 0 & 0 & \textcircled{1} & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 \quad x_3$

Solution

x_1 and x_3 are pivot columns, hence $\text{rank } A = 2$
and $\text{nullity} = \text{number of columns} - \text{rank} = 3$.

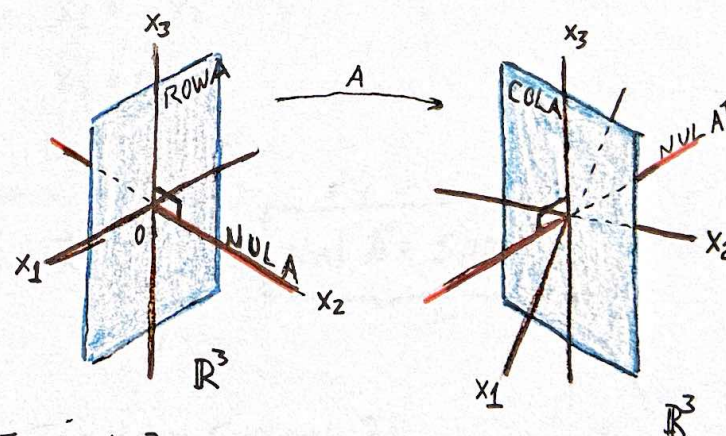
Example 6

- If A is a 7×9 matrix with nullity 2, what is the rank of A^2 ?
- could a 6×9 matrix have nullity 2?

Solution

- Since A has 9 columns, $(\text{rank } A) + 2 = 9$. Hence $\text{rank } A = 7$.
- No. If a 6×9 matrix, call it B , had a two-dimensional null space, it would have to have rank 7, by the rank theorem. But the columns of B are vectors in $\mathbb{R}^6$, and so the dimension of $\text{Col } B$ (and with that, $\text{rank } B$) cannot exceed 6.

The next example provides a nice way to visualize the subspaces we have been studying. In chapter 6 we will learn that $\text{Row } A$ and $\text{Nul } A$ have only the zero vector in common and are actually perpendicular to each other. The same fact applies to $\text{Row } A^T (= \text{Col } A)$ and $\text{Nul } A^T$. So Figure 2, which accompanies Example 7, creates a good mental image for the general case.



Example 7

Let $A = \begin{bmatrix} 3 & 0 & -1 \\ 3 & 0 & -1 \\ 4 & 0 & 5 \end{bmatrix}$. It is readily checked that $\text{Nul } A$ is the x_2 -axis.

$\text{Row } A$ is the x_1x_3 -plane, $\text{Col } A$ is the plane whose equation is $x_1 - x_2 = 0$, and $\text{Nul } A^T$ is the set of all multiples of $(1, -1, 0)$. Figure 2 shows $\text{Nul } A$ and $\text{Row } A$ in the domain of the linear transformation $\vec{x} \mapsto A\vec{x}$; the range of this mapping, $\text{Col } A$, is shown in a separate copy of $\mathbb{R}^3$, along with $\text{Nul } A^T$.

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V

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 3 & 0 & -1 \\ 4 & 0 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

$x_1 \quad x_2 \quad x_3$

$x_1, x_3 = \text{pivot}$

$x_2 = \text{free}$

$$\text{Col } A = \text{Span} \left\{ \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 5 \end{bmatrix} \right\}$$

Nul A

Solving $R\vec{x} = \vec{0}$

$$x_1 = 0$$

$$x_3 = 0$$

$$x_2 = \text{free} = s$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Let } x_2 = s$$

$$\vec{x} = s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \text{Nul } A \text{ is a line along the } x_2\text{-axis.}$$

$$A^T = \begin{bmatrix} 3 & 3 & 4 \\ 0 & 0 & 0 \\ -1 & -1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

$x_1 \quad x_2 \quad x_3$

Nul A^T

$x_1, x_3 = \text{pivots}$

$x_2 = \text{free}$

$$x_1 = -x_2 \sim x_2 = -x_1$$

$$x_3 = 0$$

$$\text{Let } x_2 = s$$

$$\vec{x} = s \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}_{\vec{v}}$$

$$\text{Nul } A^T = \text{Span}\{\vec{v}\}$$

To sum up Example 7:

- Nul A is perpendicular to Row A
- Nul A^T is perpendicular to Col A

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VI

Example 8

A scientist has found two solutions to a homogeneous system of 40 equations in 42 variables. The two solutions are not multiples, and all other solutions can be constructed by adding together appropriate multiples of these two solutions. Can the scientist be certain that an associated nonhomogeneous system (with the same coefficients) has a solution?

Solution

Yes, he can. Let A be the 40×42 coefficient matrix. The given information implies that the two solutions are linearly independent and span $\text{Nul } A$. So nullity $A = 2$. By the rank theorem, $\text{rank } A = 42 - 2 = 40$. Since $\mathbb{R}^{40}$ is the only subspace of $\mathbb{R}^{40}$ whose dimension is 40, $\text{col } A$ must be all of $\mathbb{R}^{40}$. This means that every nonhomogeneous equation $A\vec{x} = \vec{b}$ has a solution.